DAA Notes

Dijkstra's Algorithm

Dijkstra's algorithm finds the shortest path from a source vertex to all other vertices in a graph with non-negative edge weights. It keeps a distance array, repeatedly picks the unvisited vertex with the smallest distance, and relaxes its outgoing edges. With a min-heap its time complexity is O((V + E) log V).

Merge Sort

Merge sort is a divide and conquer algorithm. It splits the array into halves, sorts each half recursively and merges the sorted halves. Time complexity is O(n log n) in all cases and it needs O(n) extra space.

Dynamic Programming

Dynamic programming solves problems with overlapping subproblems and optimal substructure by storing results of subproblems. Examples are the knapsack problem, longest common subsequence and Fibonacci numbers.